

PostgreSQL Practice

15 SQL Query Challenges - Easy to Hard

Tables used - The questions are based on the provided **orders** and **products** CSV tables. The common key is **product_id**.

orders	products
order_id	product_id
product_id	product_name
quantity	category
order_date	price
customer_name	stock_quantity
payment_method	is_available
	added_on

Level 1 - Easy

1. Basic filtering

Find all orders made by the customer **Rohan**.

Return: order_id, customer_name, product_id, quantity, order_date.

2. Sorting + filtering

Find all products whose **price** is greater than **500**.

Display: product_id, product_name, category, price. Sort from highest price to lowest price.

3. Multiple conditions

Find all products that are currently available (**is_available = true**) and have **stock_quantity** greater than **10**.

Display the product name, category, stock quantity and price.

4. Date filtering

Find all orders placed between **June 20, 2025** and **June 25, 2025**, inclusive.

Display: order_id, customer_name, order_date, quantity. Sort by order_date ascending.

Level 2 - Easy / Medium

5. DISTINCT + filtering

Find all the **different payment methods** used in the orders table. Then find how many different customers have placed orders.

Try solving both using the appropriate **DISTINCT** / aggregate concepts.

6. Aggregation

Find the **total quantity of products ordered** across all orders.

Return one value: **total_quantity**.

7. GROUP BY

Find the **total quantity ordered by each customer**.

Return: **customer_name, total_quantity**. Sort from highest total quantity to lowest.

8. GROUP BY + COUNT

Find how many orders each customer has placed.

Return: **customer_name, number_of_orders**. Sort by **number_of_orders** descending.

Level 3 - Medium

9. JOIN

Using both tables, display every order along with the actual product name.

Return: **order_id, customer_name, product_name, quantity, order_date**. You will need an appropriate JOIN.

10. Calculate order value

Using both tables, calculate the value of each order.

Order Value = quantity x price. Return: **order_id, customer_name, product_name, quantity, price, order_value**. Calculate **order_value** inside the query.

11. Customer spending

Find the **total amount spent by each customer**.

Join orders with products, calculate each order value, then group by customer. Return: **customer_name, total_spent**. Sort highest spender first.

12. Category performance

Find the **total revenue generated by each product category**.

Return: **category, total_revenue**. This combines JOIN + calculated column + GROUP BY + SUM + ORDER BY.

Level 4 - Hard

13. HAVING challenge

Find customers whose **total spending is greater than Rs. 2,000**.

Return: **customer_name, total_spent**. **Restriction:** You must use HAVING for the spending condition. Do not solve it using WHERE.

14. Best-selling product

Find the product that has had the **highest total quantity sold**.

Return: **product_id, product_name, total_quantity_sold**. Think about JOIN, GROUP BY, SUM(), ORDER BY and LIMIT.

15. Final Challenge - Product Performance Report

Create a report showing **every product**, including products that have **never been ordered**.

Return: product_id, product_name, category, price, stock_quantity, total_quantity_sold, total_revenue. If a product has never been ordered, totals must show **0**, not NULL. Sort by total_revenue from highest to lowest.

Recommended progression

Questions	Main concepts
1-4	SELECT, WHERE, ORDER BY, dates
5-8	DISTINCT, COUNT, SUM, GROUP BY
9-10	JOIN, calculated columns
11-12	JOIN + aggregation
13	HAVING
14	Aggregation + ORDER BY + LIMIT
15	LEFT JOIN + COALESCE + aggregation